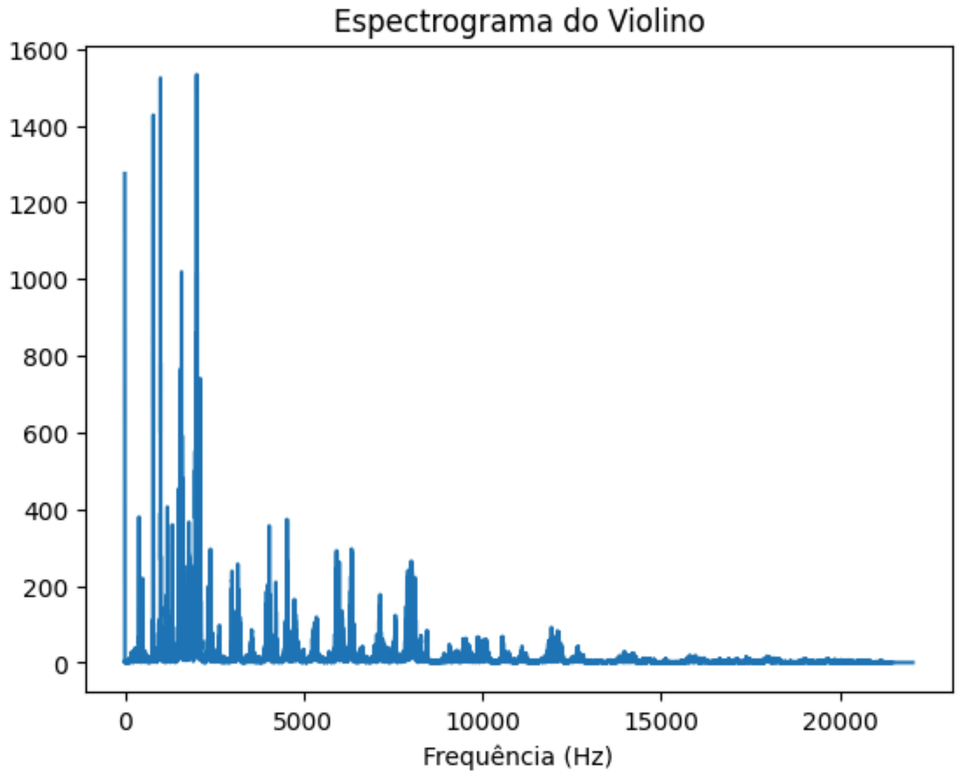
Classificador de Instrumentos Musicais

Adriel Ferreira & Kamily Assis

Motivação

Como classificação está ligada a PDI?

Espectrogramas são imagens



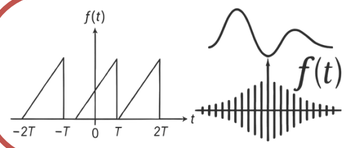
Dataset e Pipeline

kaggle

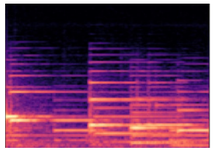
IRMAS Dataset

Áudio

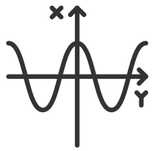
STFT



Mel



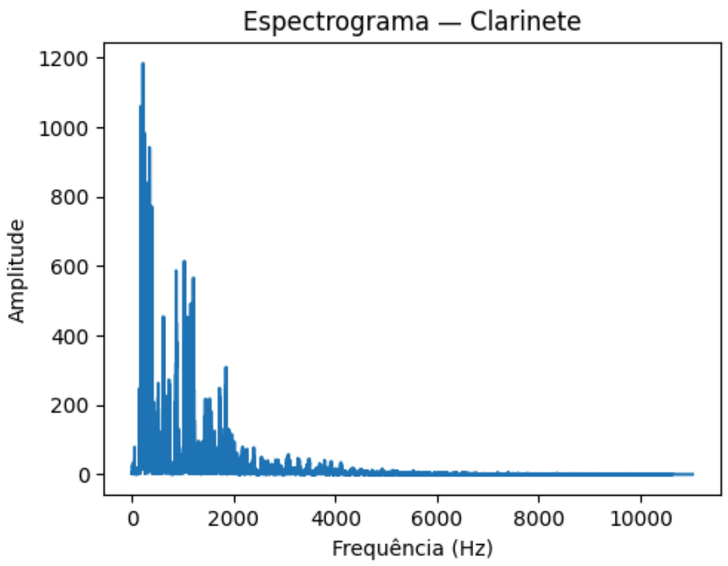
MFCC (DCT)

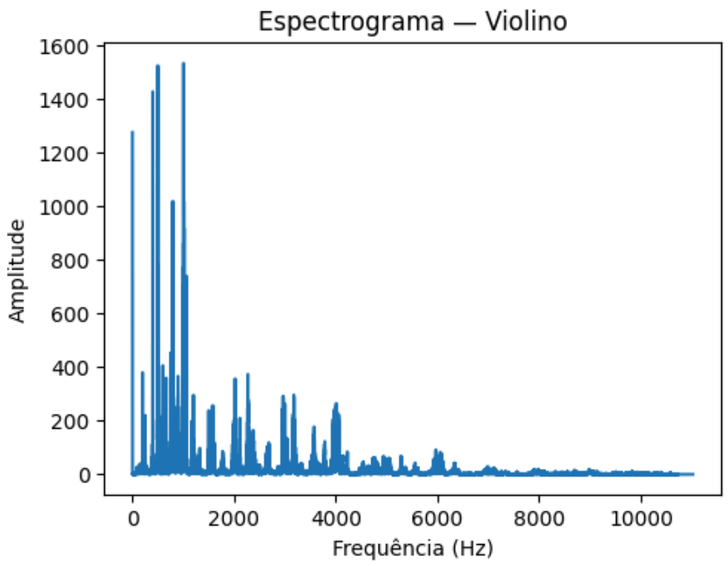


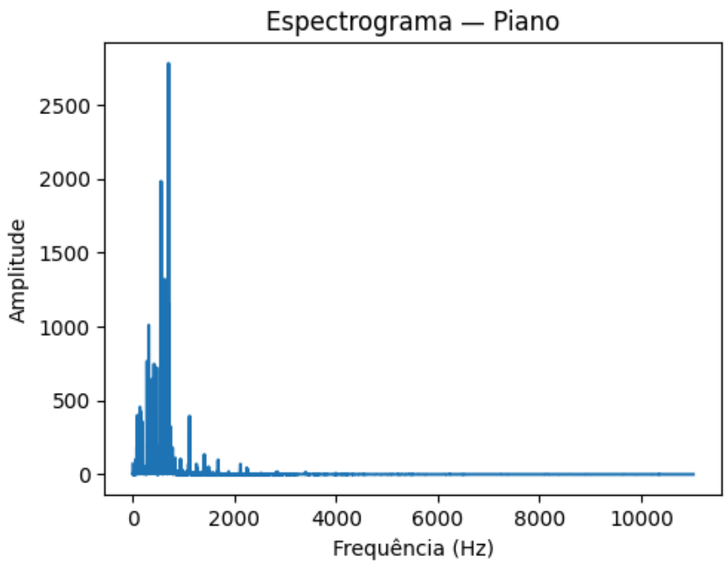
SVM

Classificação

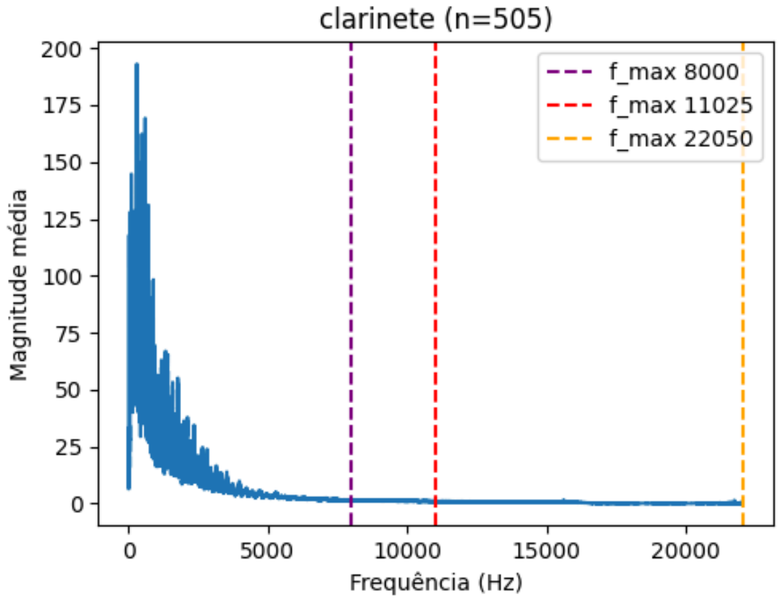
Espectrogramas

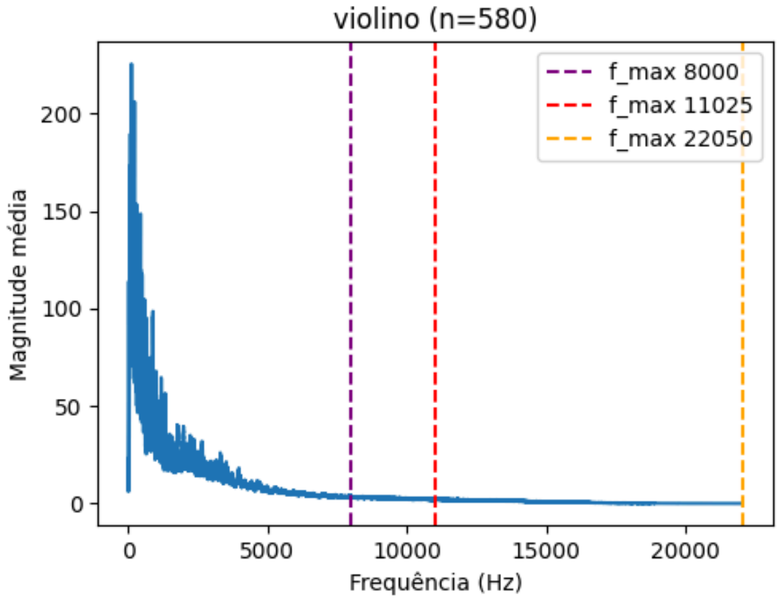


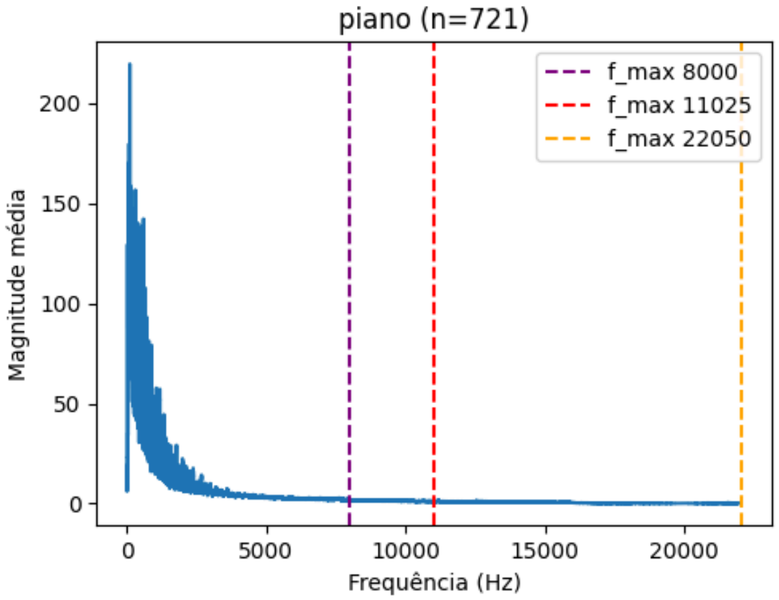




Amostragem ♫ Nyquist





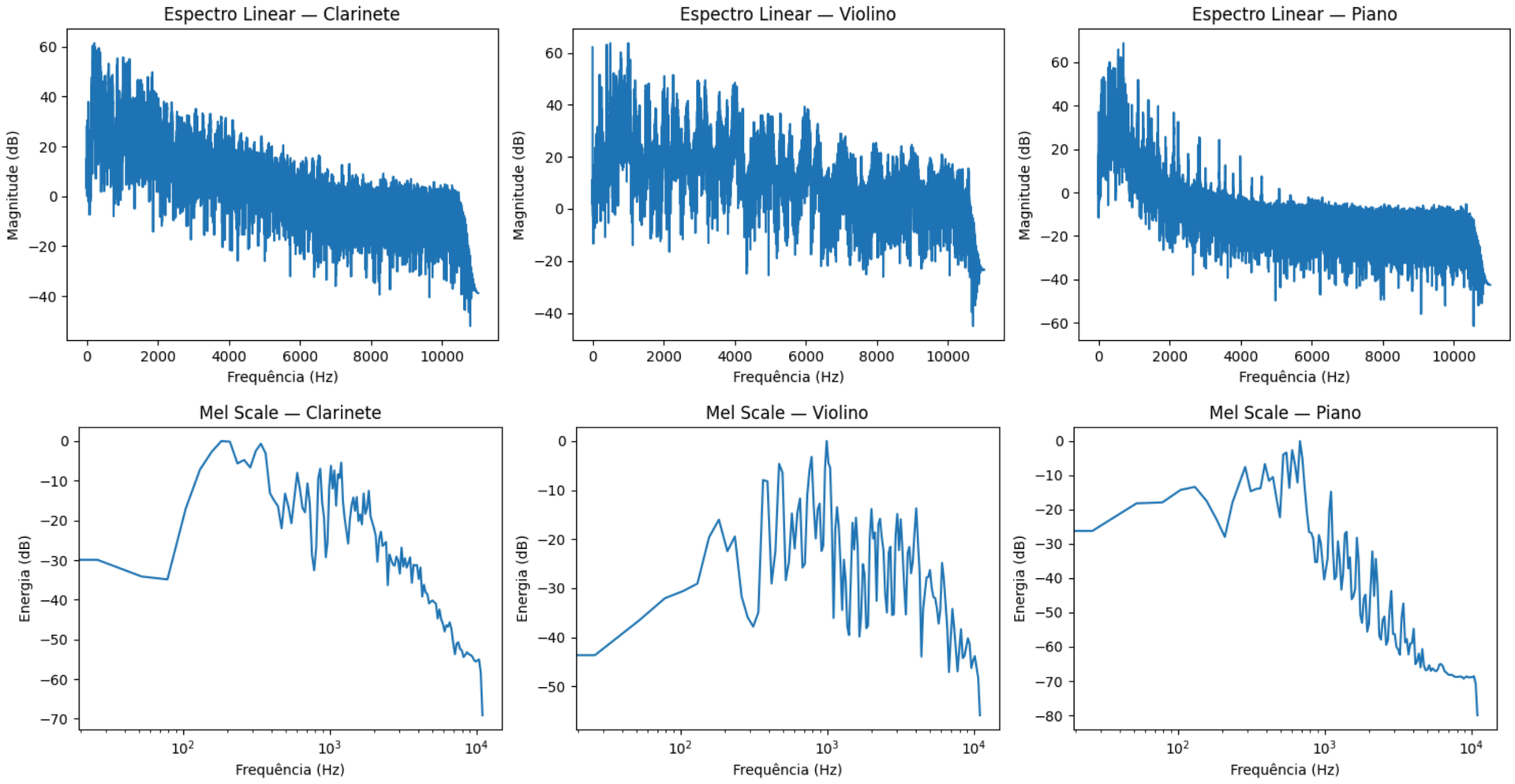


anti-aliasing: **resampy**

sr=22050

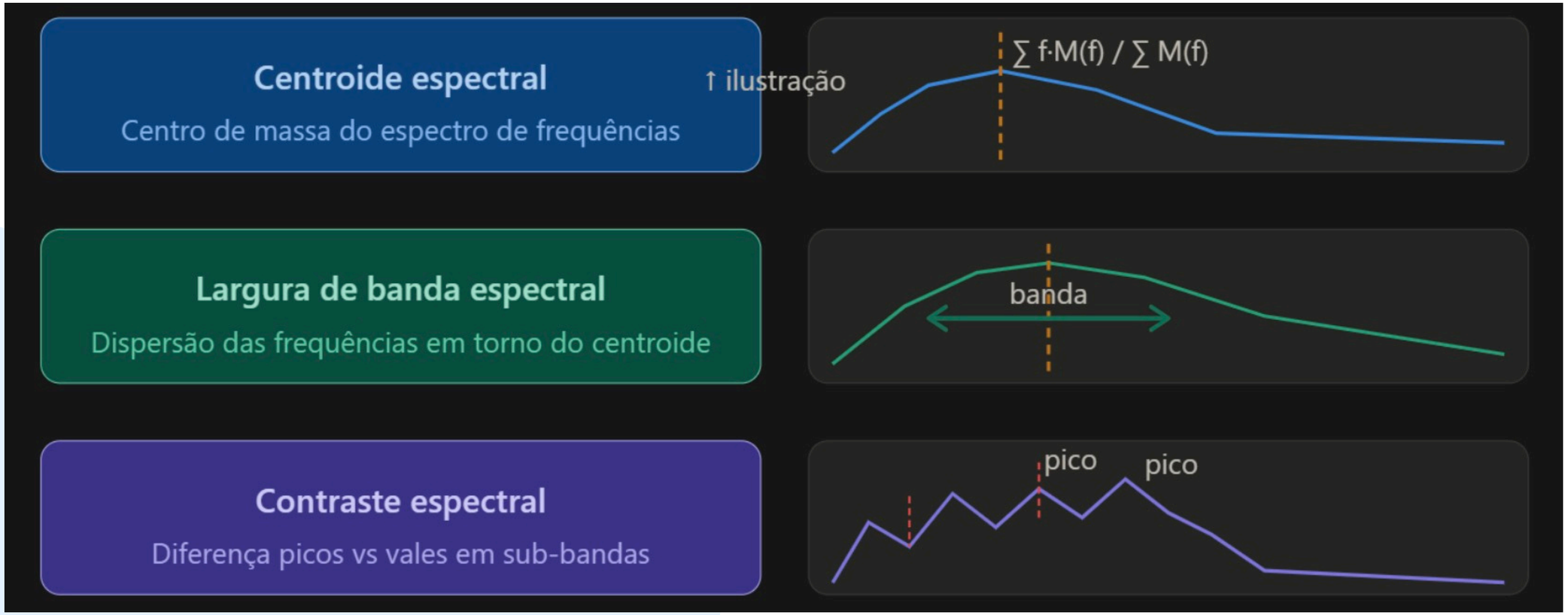
sr > 2 × f_max

Escala Mel


Espectro Linear — Clarinete
Espectro Linear — Violino
Espectro Linear — Piano
Magnitude (dB)
Frequência (Hz)
Mel Scale — Clarinete
Mel Scale — Violino
Mel Scale — Piano
Energia (dB)
Frequência (Hz)

Features de extração de áudio

6 features x 2 (média e desvio padrão)+ 13 MFCCs = 38 dimensões por áudio



Features de extração de áudio

6 features x 2 (média e desvio padrão)+ 13 MFCCs = 38 dimensões por áudio

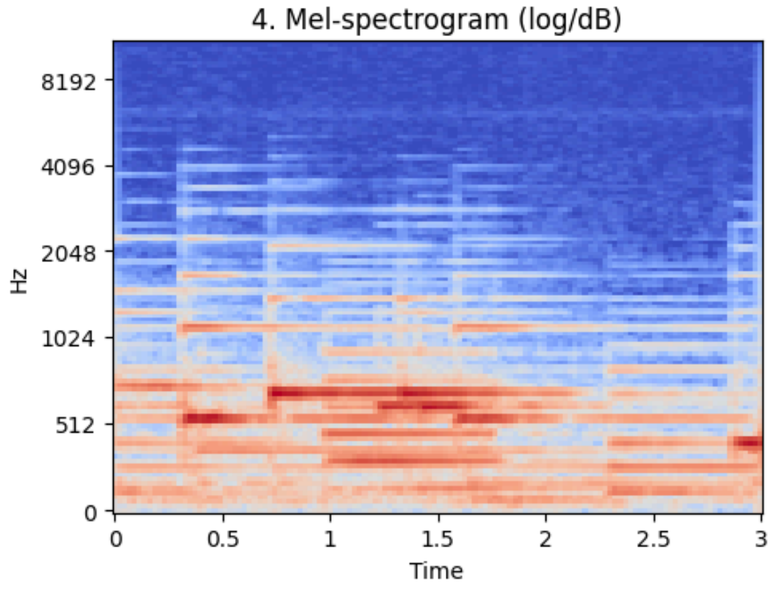
MFCC

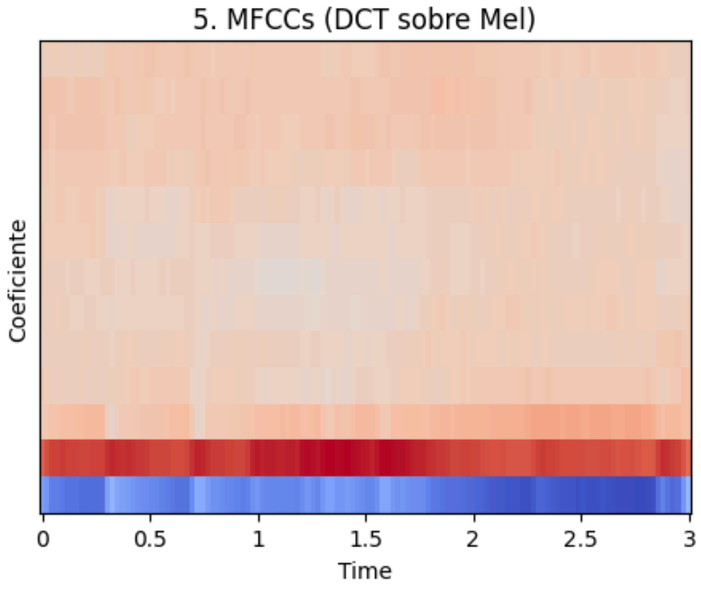
n_fft = 2048
resolução espectral

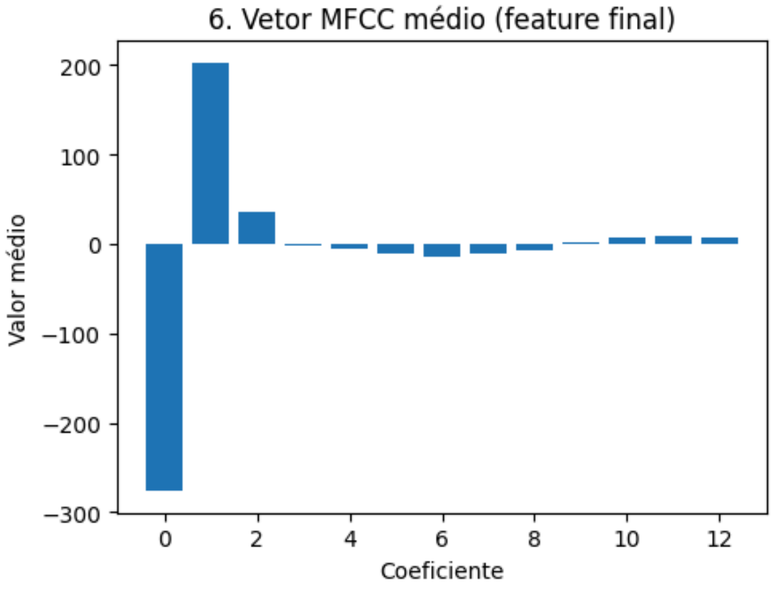
hop_lenght = 512
resolução temporal

n_mels = 128
nº faixas mel

mfcc_dim= 13
corte dos coeficientes







modelo SVM

```
── Relatório de classificação ──────────────────────
              precision    recall  f1-score   support

     Violino       0.84      0.88      0.86       116
       Piano       0.87      0.88      0.88       145
   Clarinete       0.83      0.76      0.79       101

    accuracy                           0.85       362
   macro avg       0.84      0.84      0.84       362
weighted avg       0.85      0.85      0.85       362
```

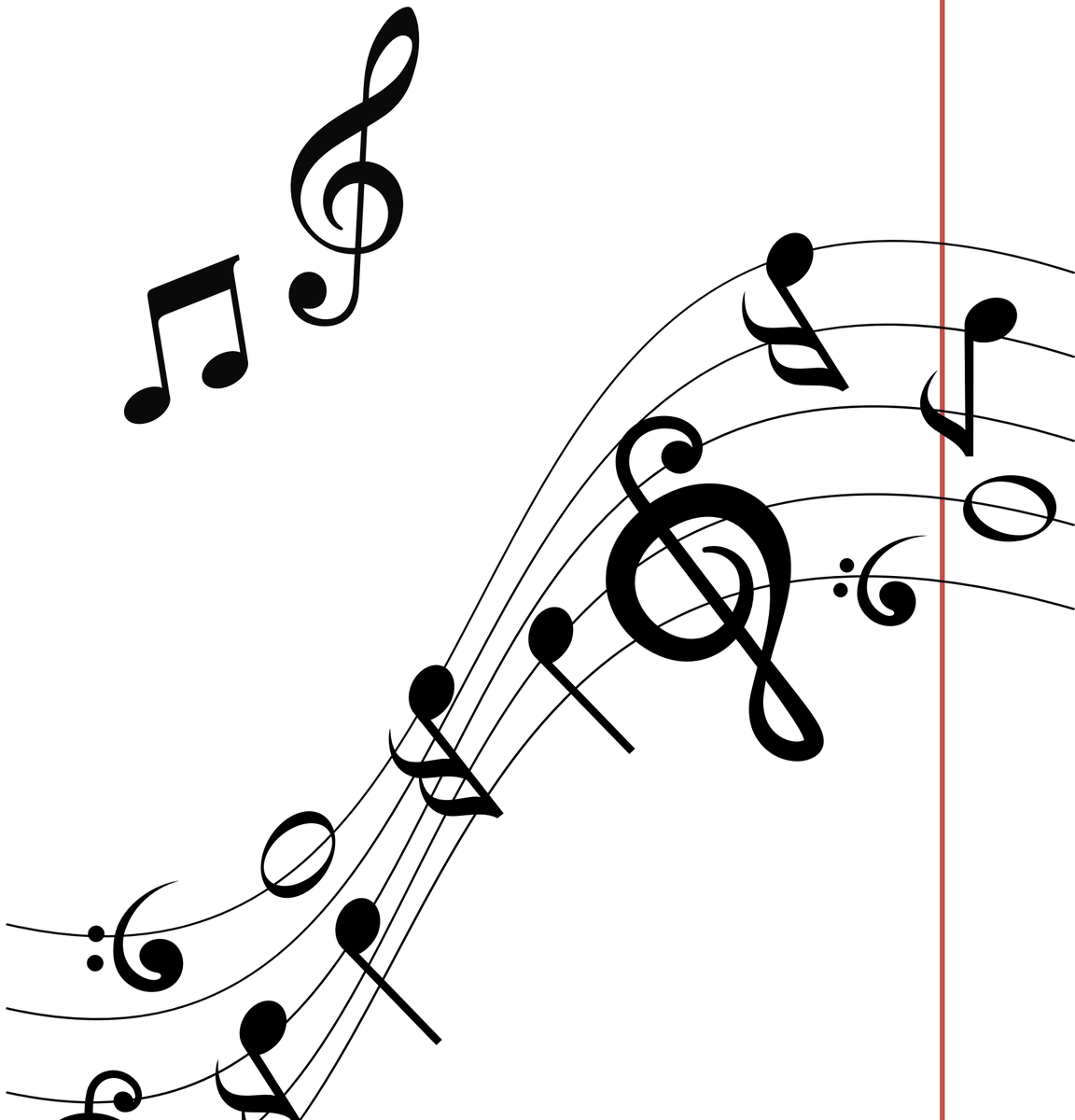
Bom desempenho
Em datasets pequenos

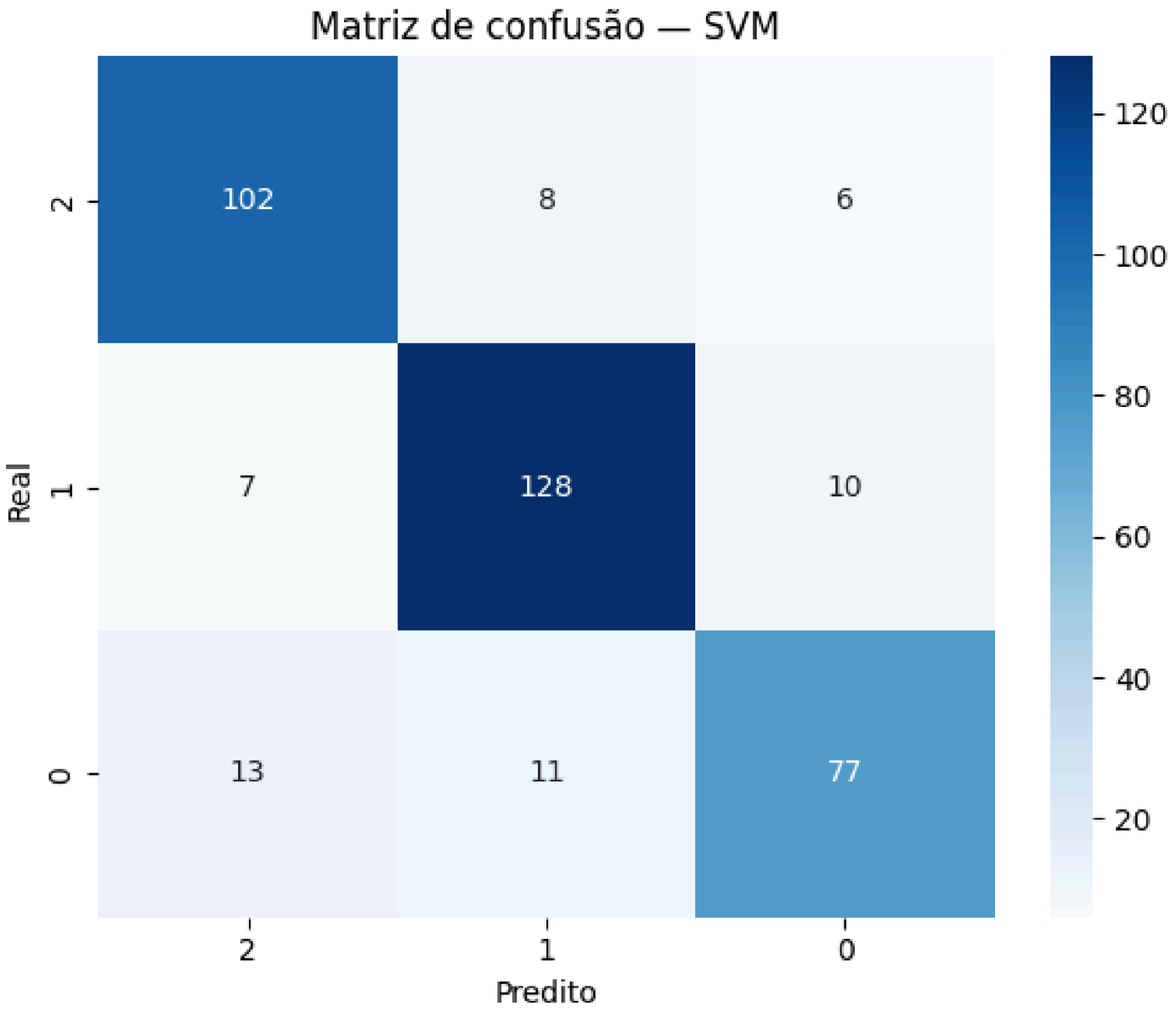
Robusto para Dimensões
MFCCs + espect.

Gratuito
Livre para uso

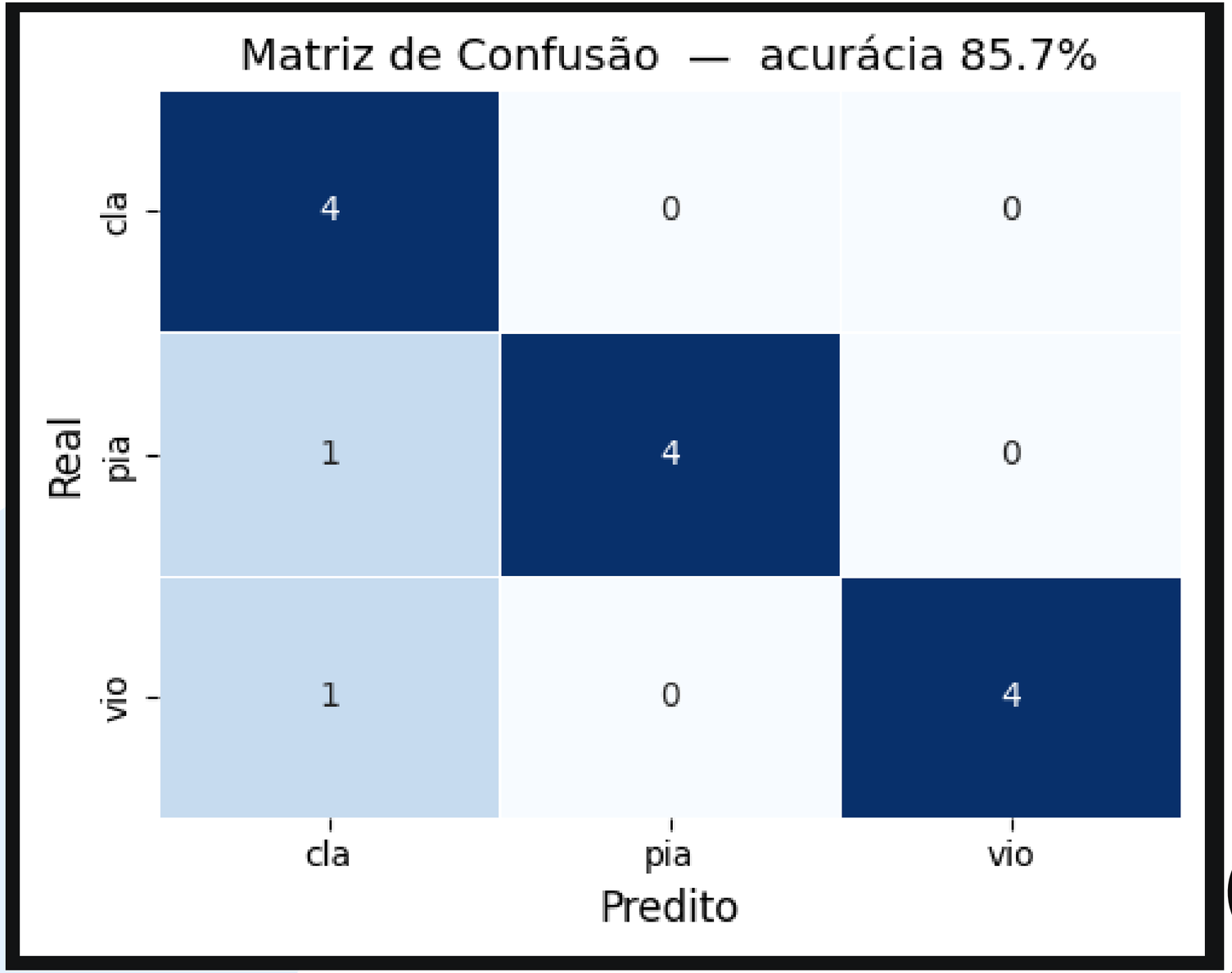
Referência Academica
para dados não-linearmente separaveis

Aúdios Mistos




Matriz de confusão — SVM
Real
Predito
2
1
0
102
8
6
7
128
10
13
11
77
120
100
80
60
40
20

Novos Dados


Matriz de Confusão — acurácia 85.7%
Real
cla
pia
vio
4
0
0
1
4
0
1
0
4
cla
pia
vio
Predito

Obrigada!